

교육용 자료 안내

본 자료는 Darun Color Institute의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

Material Safety Data Sheet (MSDS)

1. Chemical Product and Company Identification

Item	Information
Product Name	Educational Color X-RED
Recommended Use	Cosmetic ingredient
INCI Name	Polyglyceryl-2 Triisostearate, CI 77491, Cetyl Ethylhexanoate, Caprylic/Capric Triglyceride

2. Hazards Identification

Category	Details
GHS Classification	Health: 0 / Flammability: 0 / Reactivity: 0
Hazard Statements	Not Available
Precautionary Statements	P201: Please obtain instruction manual before use P281: Wear suitable personal protective equipment P410+404: Avoid direct sunlight, and store the container tightly closed P501: Dispose of the contents/container in accordance with relevant laws and regulations

3. Composition/Information on Ingredients

**Composition based on each ingredient's GHS MSDS information (KOSHA)*

Ingredient Name	CAS No.	Approximate Content (%)
Polyglyceryl-2 Triisostearate	73296-86-3	28~35%
CI 77491	1309-37-1	45~55%
Cetyl Ethylhexanoate (CEH)	59130-69-7	5~10%
Caprylic/Capric Triglyceride (MCT)	73296-86-3	5~10%

4. First Aid Measures

Route of Exposure	First Aid Procedure
Eye Contact	In case of contact with substance, immediately flush eyes with running water for at least 20 minutes. Seek medical attention immediately.
Skin Contact	In case of contact with substance, immediately flush skin with running water for at least 20 minutes. Remove and isolate contaminated clothing. Seek medical attention immediately.

Inhalation	Move to a place with fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Seek emergency medical attention.
Ingestion	Do not give anything by mouth to an unconscious person. Seek medical attention immediately.

5. Fire-Fighting Measures

Item	Information
Suitable Extinguishing Media	Small fire: dry sand, dry chemical, alcohol resistant foam, water spray, general foam, CO2 Large fire: water spray/mist, general foam
Specific Hazards	Can be ignited by heat, sparks or flames. Container may explode when heated. Fire may generate irritating or toxic gases.
Special Protective Equipment	Move container from fire area if not dangerous. In case of tank fire, cool containers with plenty of water even after extinguishing.

6. Accidental Release Measures

Action	Description
Personal Precautions	Eliminate all sources of ignition. Stop leak if not dangerous. Ventilate the contaminated area. Do not touch or walk around exposed objects. Prevent dust formation.
Environmental Protection	Prevent entry into waterways, sewers, basements and confined spaces.
Methods and Clean Up	Small spills: flush with large amount of water, absorb with sand or non-combustible material. Large spills: make a ditch away from liquid spills.

7. Handling and Storage

Category	Requirements
Handling	Be aware of substances and conditions to avoid. Wash thoroughly after handling. Work with reference to engineering controls and personal protective equipment. Prevent dust formation. Beware of high temperatures.
Storage	Keep tightly closed. Store in a cool dry place. Be aware of substances and conditions to avoid.

8. Exposure Controls/Personal Protection

Control Type	Requirement
Engineering Controls	Process isolation, use local exhaust, or keep the air level below the exposure limit.
Respiratory Protection	Wear respiratory protection certified by OSHA for the physicochemical properties of exposed particulate matter. (CI 77491: TWA 5mg/m ³)
Eye Protection	No data
Hand Protection	Wear appropriate protective gloves considering the physical and chemical properties of chemicals.
Body Protection	Wear appropriate protective clothing considering the physical and chemical properties of chemical substances.

9. Physical and Chemical Properties

Property	Value
Appearance	Red Paste
Odor	Odorless
pH	No data
Flash Point	No data
Viscosity	No data
Solubility	No data

10. Stability and Reactivity

Item	Information
Chemical Stability	Stable under normal temperature and pressure conditions
Conditions to Avoid	Ignition sources such as heat, sparks, and flames
Incompatible Materials	Combustible material
Hazardous Decomposition	Irritating, toxic gas

11. Toxicological Information

Effect	Description
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Acute Toxicity	Oral (CI 77491): LD50 > 10000 mg/kg Rat Inhalation (CI 77491): Dust LC50 5.05 mg/L 4hr Rat
Skin Irritation	CI 77491: Erythema Point 0/8, No irritation, Rabbit (OECD TG 404)
Eye Irritation	CI 77491: No irritation, Rabbit (OECD TG 404)
Respiratory Irritation	No data
Long-term Exposure	No data

12. Ecological Information

Category	Information
Ecological Toxicity	Fish (CI 77491): LC0 > 50000 mg/L 96hr (Danio rerio) Crustacean (CI 77491): EC50 > 100 mg/L 48hr
Persistency and Degradability	No data
Bio-condensability	No data

13. Disposal Considerations

Item	Information
Disposal Method	Dispose of contents and container in accordance with regulations specified in the Waste Management Act.

14. Transport Information

Item	Information
Classification	NON-HAZARDOUS FOR AIR TRANSPORT AGAINST IATA REGULATION
UN Number	Not applicable
Proper Shipping Name	Not applicable

15. Regulatory Information

Item	Information
Regulatory Basis	This GHS MSDS is based on each ingredient's GHS MSDS information from KOSHA (Korea Occupational Safety and Health Agency), modified under Article 41 of Occupational Safety and Health Acts.

16. Other Information

Item	Information
Disclaimer	To the best of our knowledge, the information contained herein is accurate. However, we cannot accept any responsibility or liability for any consequences arising from use of it. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution.
Prepared by	CHEMLAND / Darun Color Institute
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